



MASENO UNIVERSITY
UNIVERSITY EXAMINATIONS 2024/2025

**THIRD YEAR FIRST SEMESTER EXAMINATION FOR THE
DEGREE OF BACHELOR OF SCIENCE IN AGRICULTURAL
ECONOMICS AND BACHELOR OF SCIENCE IN
AGRIBUSINESS MANAGEMENT WITH
INFORMATION TECHNOLOGY**

MAIN CAMPUS

AEG 301: AGRICULTURAL PRICE ANALYSIS

Date: 21st January, 2025

Time: 3.30 - 6.30pm

INSTRUCTIONS:

- Do not write anywhere on this Question paper
- Answer ALL Questions in Section A and any other TWO in Section B.

SECTION A (COMPULSORY) (40 MARKS)

Question One

- a) Write short notes on the following terms as used in agricultural price analysis.
- i) Law of one price. (2 Marks)
 - ii) Border price. (2 Marks)
 - iii) Marketing margin. (2 Marks)
 - iv) Consumer surplus. (2 Marks)
- b) Kenya has been importing Milk from Uganda, however, Dairy farmers in Kenya have petitioned the government to restrict milk importation after suffering huge losses. The cabinet secretary for trade want to impose an import tariff on this commodity. Explain the welfare implications of this directive from the cabinet secretary, using a diagram for illustration. (7 Marks)
- c) The government of Kenya has recently reviewed the country's maize consumer price. Following the advice of the country's Chief Price Policy Advisor, the reviewed price was benchmarked on border prices. Consumers have however complained that the price is too high for them. The complaints have raised political concern and the government is considering providing a consumer

subsidy to address the situation. Using a well-labeled diagram, show and explain the impacts of the proposed subsidy. (8 Marks)

d) Using a diagram demonstrate how utility maximization can be translated into individual response or demand curve for a given product. (7 Marks)

e) Coffee farmers have been complaining of low prices from their exports. The Ministry of Trade is considering providing farmers with a positive producer subsidy. With the use of a well-labelled diagram, discuss the production, consumption, budget, and efficiency effect of this policy on the exportable commodity. (10 Marks)

SECTION B (Answer TWO questions) (30 MARKS)

Question Two

a) F. Gasmi, J.-J. Laffont, and Q. Vuong in the Journal of Economics and Management Strategy, Summer 1992 estimated the following demand functions for Coke and Pepsi:

$$Q_c = 26.17 - 3.98P_c + 2.25P_p + 2.60(A_c)^{0.5} - 0.62(A_p)^{0.5} + 0.99I + 9.58S$$

$$Q_p = 17.48 + 1.40P_c - 5.48P_p - 4.81(A_c)^{0.5} + 2.83(A_p)^{0.5} + 1.92I + 11.98S$$

The symbols are as follows:

Q_c = quantity of Coke per quarter (units of ten million cases).

Q_p = quantity of Pepsi per quarter (units of ten million cases).

P_c = price of Coke (1986 dollars per ten cases)

P_p = price of Pepsi (1986 dollars per ten cases)

A_c = quarterly advertising expenditure by Coke (millions of 1986 dollars)

A_p = quarterly advertising expenditure by Pepsi (millions of 1986 dollars)

I = average per capita disposable income (thousands of 1986 dollars)

S = "Dummy variable" = 1 for summer and spring quarters, 0 for fall and winter

The average values of the independent variables in the data were as follows:

$$P_c = 12.96, P_p = 8.16, A_c = 34.69, A_p = 27.88, I = 20.63$$

Where: P_c is the average price of Coke, P_p is the average price of Pepsi, A_c is the average quantity of Coke, A_p is the average quantity of Pepsi and I is the average income.

Consider the case of spring and summer seasons ($S = 1$). **Required:** Calculate and interpret;

- i) Own price elasticity for Coke and Pepsi. (3 Marks)
 - ii) Cross-price elasticity for coke and Pepsi. (3 Marks)
 - iii) Income elasticity for coke and Pepsi. (3 Marks)
 - iv) Own advertising expenditure for coke and Pepsi. (3 Marks)
- b) Prove that elasticity is a unitless measure of responsiveness of demand/supply to price changes. (3 Marks)

Question Three

- a) A certain small developing country has been very dependent on imports for its rice consumption requirements. The government is however considering an import substitution policy aimed at expanding local production of rice. As the country's Chief Price Policy Analyst, it is your duty to advise the government on the merits and demerits of the proposed policy. To facilitate your task, you have been provided with the information below by the ministries of trade and agriculture.

FOB port of export	US\$200/Metric ton (MT)
Freight to port of import	US\$30/Metric ton (MT)
Transport and handling cost to the country's major area of consumption	KShs. 3000/Metric ton (MT)

Transport and handling cost from the country's major paddy production area to the major consumption area	KShs. 2000/Metric ton (MT)
Processing conversion factor	1 ton paddy rice; 0.64 processed rice
Cost of rice milling	KShs. 2 per Kg of paddy
Exchange rate	KShs. 75: 1 US\$
Estimated domestic farm production cost	KShs. 35/Kg paddy

Showing all your computation, analyze the policy and give your recommendation.

(9 Marks)

- b) Discuss the method of designing price policy interventions to benefit the targeted group of people.

(6 Marks)

Question Four

- a) The government of Kenya is considering introducing price control on a variety of consumer goods being sold in the Country. As a price analyst expert, using a diagram advise the government on the implications of introducing price ceiling.

(5 Marks)

- b) Explain how macroeconomic policy can affect farm prices in Kenya.

(5 Marks)

- c) Explain the main rationale for government intervention in agricultural sectors.

(5 Marks)

Question Five

- a) Explain the impact of each pricing policy adopted in Kenya **BEFORE** and **AFTER** the 1980s on the prices of the major agricultural commodities and the overall performance of the agricultural sector.

(8 Marks)

b) The following is an estimated global demand function for coffee:

$Q_c = 100.6125 - 0.393P_c$. Assume that the average consumption of coffee over ten years was 61.33 million Kilograms and the average price was Ksh 74.54 per kilogram. Determine the point elasticity of demand. (4 Marks)

c) Explain the roles played by the futures markets in the economy. (3 Marks)
